

Exercise 02: Hotel Room

Design a `HotelRoom` class for a hotel management system.

- A room has a room number, price per night, and availability status.
- The availability status must not be changed directly from outside the class.

The room should support:

- Book — book the room if it is currently available.
- Checkout — make the room available again after the guest leaves.
- An already booked room cannot be booked again.
- An available room cannot be checked out.
- The current availability status should be readable from outside the class.

Design the class so that a `HotelRoom` object can protect itself from invalid state transitions.

What Is Expected from You

In this practice, you are expected to demonstrate your ability to:

- Design a class with appropriate data and behaviors.
- Follow proper naming conventions.
- Provide multiple ways to initialize an object when appropriate.
- Keep the internal state (data) private.
- Apply encapsulation properly—not just getters and setters.
- Ensure the object protects its own state, so callers cannot put it into an invalid state.